Multimodal Large Model+video understanding（Voice Version）

Before running the function, you need to close the App and large programs. For the closing method, refer to [4. Preparation] - [1. Manage APP control services].

1. Function Description

After the program runs, input questions about video through the terminal. The large model will first record a video, then analyze the video content with the question, and finally reply with the answer to the question. You can also first record a video of specified duration, then ask the large model questions related to the video based on the video content.

2. Startup

Users with Jetson-Nano mainboard version need to enter the docker container first and then input the following command. Users with Orin mainboard can directly open the terminal and input the following command:

```
ros2 launch largemodel largemodel_control.launch.py
```

After waking up the module, give video recording commands. You can refer to the following example:

```
Record a 20-second video
```

After the buzzer beeps once,

```
[asr-5] [INFO] [1775201269.036976523] [asr]: -
[asr-5] [INFO] [1775201270.835488877] [asr]: XunFei ASR Finish.
[asr-5] [INFO] [1775201270.836123714] [asr]:  Record a 20 seconds video.
[asr-5] [INFO] [1775201270.836783532] [asr]: 😊okay, let me think for a moment...
[model_service-3] [INFO] [1775201278.987083290] [model_service]: Decision making AI planning:1. Call the `record_video` function, pass in the parameter "20"
, and record a 20-second video.
[model_service-3] [INFO] [1775201282.063814184] [model_service]: "json_str": {"response": "Okay, I'm going to record a 20-second video now. Let's see what h
appens!", "action": ["record_video(20)"]}
[model_service-3] [INFO] [1775201282.064469446] [model_service]: "action": ['record_video(20)'], "response": Okay, I'm going to record a 20-second video now
. Let's see what happens!
[action_service-4] [INFO] [1775201282.072589173] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775201282.073058176] [action_service]: "actions": ['record_video(20)']
[action_service-4] [INFO] [1775201320.716077268] [action_service]: Published message: Robot feedback: Execution record_video() completed
[action_service-4] [INFO] [1775201320.716981124] [action_service]: Checking...
[action_service-4] [INFO] [1775201320.729934728] [action_service]: pid_num:[713]
[action_service-4] [INFO] [1775201322.747178486] [action_service]: sleep_pids:[1, 709]
[model_service-3] [INFO] [1775201323.183670923] [model_service]: "json_str": {"response": "The video has been recorded successfully! I'm ready to help with
anything else you'd like to do.", "action": ["finishtask()"]}
[model_service-3] [INFO] [1775201323.184483168] [model_service]: "action": ['finishtask()'], "response": The video has been recorded successfully! I'm ready
 to help with anything else you'd like to do.
[action_service-4] [INFO] [1775201323.194199311] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775201323.194760629] [action_service]: "actions": ['finishtask()']
```

After the buzzer beeps once, recording ends. Then based on this video content, wake up the voice module and say:

```
Did you see the red block in the video I just recorded?
```

The question here should be asked based on the actual recorded video content,

```
[asr-5] [INFO] [1775202281.808717375] [asr]: -
[asr-5] [INFO] [1775202281.868207242] [asr]: -
[asr-5] [INFO] [1775202283.726617223] [asr]: XunFei ASR Finish.
[asr-5] [INFO] [1775202283.727064576] [asr]:  Did you see the red block in the video I just recorded?
[asr-5] [INFO] [1775202283.727426391] [asr]: 😊okay, let me think for a moment...
[model_service-3] [INFO] [1775202284.316867405] [model_service]: "json_str": {"response": "I see the red block is still there, but something seems to be obstructing the view—maybe a hand or
object passed in front of the camera. Let me try to get a clearer shot!", "action": ["seewhat()"]}
[model_service-3] [INFO] [1775202284.317452684] [model_service]: "action": ['seewhat()'], "response": I see the red block is still there, but something seems to be obstructing the view—maybe
 a hand or object passed in front of the camera. Let me try to get a clearer shot!
[action_service-4] [INFO] [1775202284.326387586] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775202284.327124180] [action_service]: "actions": ['seewhat()']
[model_service-3] [INFO] [1775202294.554542308] [model_service]: "json_str": {"response": "I've reviewed the video you recorded, and I can confirm that the red block was indeed visible in it
. It appeared near the monitor on the desk. Let me know if you'd like me to do anything else with it!", "action": ["finishtask()"]}
[model_service-3] [INFO] [1775202294.555452723] [model_service]: "action": ['finishtask()'], "response": I've reviewed the video you recorded, and I can confirm that the red block was indeed
 visible in it. It appeared near the monitor on the desk. Let me know if you'd like me to do anything else with it!
[action_service-4] [INFO] [1775202294.560504379] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775202294.561081676] [action_service]: "actions": ['finishtask()']
```

Location of saved video files: Videos are saved under /root.

3. Core Code Analysis

You can refer to the content in **3. Core Code Analysis** from tutorial [18. AI Model - Text Version] - [4. Multimodal Large Model+video understanding]. The voice version and text version have the same action functions, only the task command input method is different.

4. Combination with Robotic Arm

We can also combine robotic arm + video understanding to develop new gameplay. For example, prepare several opaque paper cups. After the video starts, we place an object under one of the opaque paper cups, then slowly change the positions of the paper cups. After the video recording ends, we ask the large model which cup the object is finally under, and finally have the robotic arm point to that cup. To test this function, you can wake up the voice module and input by voice:

```
Let's play a game. First record a video, then guess which cup has the red block
at the end of the video?
```

The program will record a video, then analyze which cup contains the red block, and finally control the robotic arm to point to the cup hiding the red block.

The task planning should be:

```
1. Call the `record_video` function, pass in the parameter "20", and record a 20-
second video;

2. Call the `video_understanding` function to analyze the recorded 20-second
video, and determine which cup has the red block hidden under it at the end of
the video;

3. Call the `point_to(x1, y1, x2, y2)` function to point to the cup with the red
block, where x1, y1, x2, y2 represent the outer frame coordinates of the cup with
the red block at the end of the video.
```